PQC DAILY STUDY SERIES

PQC in Email Security: The Path to Quantum-Resistant S/MIME and PGP/GPG

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Topic of the Day

PQC in Email Security: The Path to Quantum-Resistant S/MIME and PGP/GPG.

EN Content

As the world prepares for the quantum computing era, securing our digital communications is paramount. Email, a cornerstone of our digital identity and communication, relies heavily on cryptographic standards like S/MIME (Secure/Multipurpose Internet Mail Extensions) and PGP/GPG (Pretty Good Privacy/GNU Privacy Guard). These standards, however, are built upon classical cryptographic algorithms (RSA, ECC) that are vulnerable to attacks from large-scale quantum computers. Today, we explore the critical transition to quantum-resistant email security.

The primary threat to email security is the "Harvest Now, Decrypt Later" (HNDL) attack model. Adversaries can intercept and store encrypted emails today, waiting for the day a quantum computer can break the encryption and reveal their contents. This makes the transition to Post-Quantum Cryptography (PQC) for email an immediate necessity, not a distant concern.

Applicability to Real-World Cybersecurity

The integration of PQC into email standards like S/MIME and PGP/GPG involves a multi-faceted approach:

- **Hybrid Signatures and Encryption:** The most practical near-term solution is a hybrid approach. This involves creating signatures and encrypting messages with both a classical and a PQC algorithm. For example, an email could be signed with both RSA and ML-DSA. This ensures that the email remains secure against both classical and quantum adversaries. If one algorithm is broken, the other still provides security.

- **Updating X.509 Certificates:** S/MIME relies on X.509 certificates for trust. These certificates will need to be updated to include PQC public keys and be signed by Certificate Authorities (CAs) using PQC-based digital signatures. This transition is already underway, with CAs beginning to issue hybrid certificates.
- **Client and Server-Side Support:** Email clients (like Outlook, Thunderbird) and mail servers (like Exchange, Postfix) will need to be updated to support the new PQC algorithms. This is a significant undertaking that requires coordinated efforts from software vendors and the open-source community.

Comparison of PQC in Email Security

Aspect	Classical (RSA/ECC)	Post-Quantum (ML-DSA/ML-KEM)
Security	Vulnerable to quantum attacks	Resistant to known quantum attacks
Signature Size	Relatively small	Larger, which can impact email size
Performance	Fast and efficient	Varies by algorithm, can be slower
Adoption	Widespread	In progress, requires client/server updates

Daily Quiz

1. What is the "Harvest Now, Decrypt Later" (HNDL) attack and why is it relevant to email security?
2. What is a hybrid signature and why is it a practical near-term solution for securing emails?
3. What are the two main components that need to be updated to support PQC in email?